

RECONSTRUCTING CURRENTS FROM THEIR PROJECTIONS

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ABSTRACT. We prove an inversion formula for the exterior k -plane transform. As a consequence, we show that if $m < k$ then an m -current in $\mathbf{R}^n$ can be reconstructed from its projections onto $\mathbf{R}^k$, which proves a conjecture of Solomon.

A basic problem in analysis and geometry is to reconstruct an object in euclidean space from its orthogonal projections. Here we are interested in reconstructing an oriented submanifold, or more generally a current, from its projections. The geometry needed to prove our main theorem was worked out by Solomon [Sol11], who conjectured that the analysis would work out as well. In this brief note, we prove that conjecture.

Throughout, fix integers $0 \leq m < k < n$. A *compactly supported m -current* is a continuous linear functional on the space of smooth m -forms [Sim84, §6.2]. If T is a compactly supported m -current and α is a smooth m -form, we write $\int_T \alpha$ for $T(\alpha)$. Thus every compact, oriented, C^1 submanifold of $\mathbf{R}^n$ is a compactly supported m -current.

Let G be the Grassmannian variety of k -dimensional linear subspaces of $\mathbf{R}^n$. There is a natural $O(n)$ -action on G , which induces a unique $O(n)$ -invariant Borel probability measure on G . For any $P \in G$, we have two operations:

- (1) Given an m -form α on P , the *pullback* $P^*\alpha$ is the m -form on $\mathbf{R}^n$ which is the pullback of α by the orthogonal projection, $\mathbf{R}^n \rightarrow P$.
- (2) Given a compactly supported m -current T on $\mathbf{R}^n$, the *pushforward* P_*T is the compactly supported m -current on P such that for every m -form α on P ,

$$\int_{P_*T} \alpha = \int_T P^*\alpha.$$

It is very important to work in the category of compactly supported currents for the pushforward by an orthogonal projection to make sense. Indeed, one can only push forward a noncompactly supported current by a proper map, which orthogonal projections are not; moreover, it does not really make sense to make sense to push forward a submanifold of $\mathbf{R}^n$.

We take the convention $0 \in \mathbf{N}$. Let $\langle x \rangle := \sqrt{1 + |x|^2}$ be the Japanese bracket. By $A \lesssim_k B$ we mean that there exists $C > 0$ (which depends on k) for every A, B , $A \leq CB$. The *Schwartz space* is the Fréchet space $\mathcal{S}$ of m -forms α such that for every $N \in \mathbf{N}$, $|\alpha(x)| \lesssim_N \langle x \rangle^{-N}$ [Hör15, §7.1]. A density argument shows that a compactly supported m -current is determined by its action on $\mathcal{S}$.

Our main theorem establishes [Sol11, Conjectures 6.5 and 6.8]:

Theorem 1. *For every $\alpha \in \mathcal{S}$ and $P \in G$, there exists a smooth m -form α_P on P such that for every $x \in \mathbf{R}^n$,*

$$\alpha(x) = \int_G P^*(\alpha_P)(x) \, dP, \tag{1}$$

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